



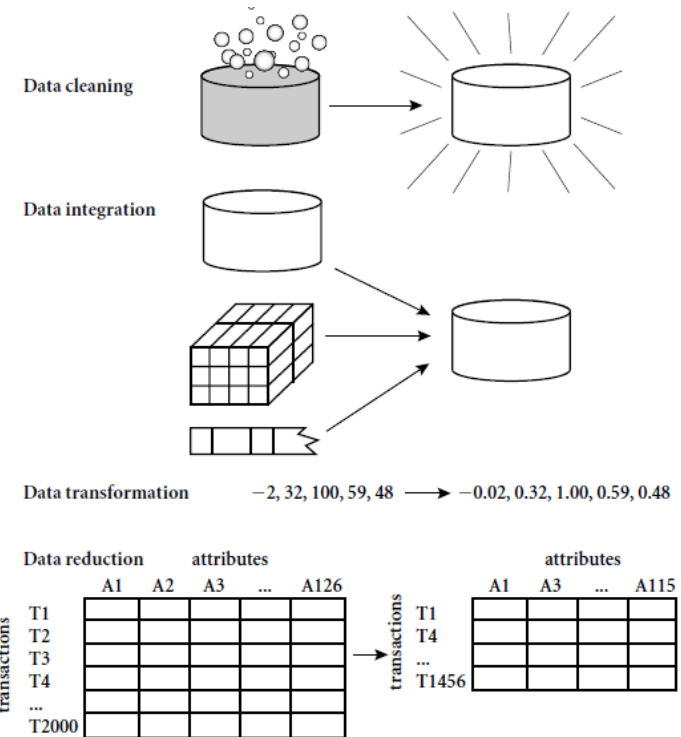
Applied Data Mining

**— Week 5—
Spring 2026**

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Outline

- Data Preprocessing: Definition, Goal
- Why do we preprocess the data?
- Major Tasks in Data Preprocessing
 - Data Cleaning
 - Data Integration
 - Data Reduction
 - Data Transformation



DATA TRANSFORMATION



A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values.

Methods

- Normalization: Scaled to fall within a smaller, specified range
- min-max normalization
- z-score normalization

NORMALIZATION (EXAMPLE)

- **Min-max normalization:** transform the values to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Given Data = 12K, 50K, 73.6K, 98K
- Ex. The values \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,600 is mapped to

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

NORMALIZATION (EXAMPLE)

- **Z-score normalization** (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

- Ex. Let $\mu = 58,400$, $\sigma = 63420$. Then

- $$\frac{73600 - 58400}{63420} = 0.239$$

- Formula: $\sigma = \sqrt{\sum (x_i - \mu)^2} = 63,420$